

## Review: Identifying similar triangles



- 1
  - a
    - i A and C
    - ii All sides are in the same ratio.
  - b
    - i B and D
    - ii All corresponding angles are congruent.
- 2 A and B
- 3 Another pair of congruent angles must be added.
- 4
  - a No
  - b Yes
  - c No
  - d No
- 5
  - a No
  - b No
- 6
  - a The triangles are similar by Angle-angle similarity ( $AA\sim$ ).
  - b The triangles are not similar. The sides are not in the same proportion.
  - c The triangles are similar by Side-angle-side similarity ( $SAS\sim$ ).
  - d The triangles are similar by Side-side-side similarity ( $SSS\sim$ ).
  - e The triangles are similar by Side-angle-side similarity ( $SAS\sim$ ).
  - f The triangles are similar by Angle-angle similarity ( $AA\sim$ ).
  - g The triangles are similar by Side-angle-side similarity ( $SAS\sim$ ).
  - h The triangles are not similar. The sides are not in the same proportion.
- 7 Each of the angles measures  $60^\circ$ . This means two equilateral triangles will always have three pairs of congruent angles, and so they will satisfy Angle-angle-angle (AAA) similarity condition.

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## Additional questions

8

a

i A and D

ii HL



i B and D

ii AA

c

i A and B

ii HL

d

i C and D

ii SSS

e

i A and C

ii SAS

f

i B and C

ii SAS

g

i B and D

ii HL

h

i B and D

ii AA

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a  $\triangle ABC$  and  $\triangle ADE$  are similar by AA

b  $\triangle JMN$  and  $\triangle JKL$  are similar by AA

10 AA

11  $\frac{AB}{BD} = \frac{AC}{DE}$

12 Example answers:

- $\frac{FX}{GX} = \frac{3}{7}$
- $EF$  and  $GH$  are parallel.

13

a The triangles don't have matching sides in the same ratio.

b  $YZ = 9$